# NFC SM Branch Successor Pilot

v30 — L2 Generated Shell

First exercise of `intrinsic-structural-close` posture

Generated governance shell

## Contents



**Governance notice.** This is a machine-generated L2 branch-pilot shell. It is *not* a supersession of the canonical NFC Standard Model Branch Book. The canonical SM branch book remains the authoritative source of truth. Full supersession requires L5 gate passage per the NFC supersession policy.

## SM Branch Successor Pilot

This generated packet is the v30 L2 branch-pilot shell for the Standard Model super-branch. It is *not* a supersession of the canonical SM Branch Book. The pilot exercises the **intrinsic-structural-close** posture for the first time: SM has closed its internal structural obligations while remaining open on inherited scope from the YM and GR sub-branches.

### Posture and inheritance structure

Computed SM posture: **intrinsic-structural-close**. This posture encodes: *(i)* conditional intrinsic structural closure (four internal obligations discharged at structural-conditional level); *(ii)* inherited-scope open (all YM and GR obligations propagate to SM); *(iii)* active frontier `ob:SM.O-IDcont` (continuum/operator identification force upgrade, the single named irreducible frontier of the SM branch).

### Status inequality

The governing constraint is: $\mathrm{Status(SM)} \leq \mathrm{Status(YM)} \wedge \mathrm{Status(GR)} \wedge \mathrm{Status}(\mathrm{SM}_{\mathrm{intrinsic}})$. The pilot preserves this inequality explicitly. SM cannot be promoted to conditional-cert-close or cert-close by any discharge internal to the SM packet.

### Non-claims (strictly maintained)

- No measured particle masses
- No CKM/PMNS mixing parameters
- No renormalization-group running derivation
- No full empirical Standard Model reconstruction
- No gauge-gravity unification claim

- No unconditional CERT-CLOSE

Table 1: SM successor-pilot claim catalog

| ID | Kind | Decl. | Comp. | Scope |
|---|---|---|---|---|
| ob:SM.harmonization | obligation | C | C | scoped-KPO |
| ob:SM.matter | obligation | C | C | scoped-candidate-assignment |
| ob:SM.higgs | obligation | C | C | scoped-B0-screening |
| ob:SM.coupling-transfer | obligation | C | C | structural-seed |
| thm:SM.gauge-chain | theorem | C | C | branch |
| thm:SM.three-gen | theorem | U | U | branch |
| thm:SM.harmonization-partial | theorem | C | C | scoped-KPO |
| thm:SM.coupling-transfer-full | theorem | C | C | structural-seed |
| thm:SM.higgs-source-descended | theorem | C | C | scoped-B0-screening |
| thm:SM.18.1 | theorem | C | C | branch |
| prop:SM.intrinsic-closure | proposition | C | C | intrinsic-structural |
| prop:SM.status-reconciliation | proposition | C | C | intrinsic-structural |
| ob:SM.O-IDcont | obligation | O | O | branch |
| status:SM | status_note | intrinsic-structural-close | intrinsic-structural-close | branch |

## Generated claim shells

### Obligation: Harmonization Functor (O-SM.Harmonization)

**Machine ID.** `ob:SM.harmonization`
**Declared.** C **Computed.** C
**Scope.** regime=SM-super-sector; domain=SM-branch; certificate_scope=scoped-KPO; promotion_ceiling=C

Construct the SM harmonization functor F_SM: it must take the pair (E_YM, E_GR) of sub-branch endpoint data and produce a six-component SM endpoint datum (gauge algebra, mass gap, metric class, gravitational coupling, cosmological constant, normalization unit) satisfying source-descendedness, no hidden supplementation, and the Book VII scoped-certificate condition. The KPO chain (KPO-1 through KPO-4) is the canonical candidate; extending to the full matter sector requires O-SM.Matter and O-SM.Higgs at unconditional force.

*Dependencies.* thm:YM.7, thm:GR.CP.1

### Obligation: Matter Content (O-SM.Matter)

**Machine ID.** `ob:SM.matter`
**Declared.** C **Computed.** C
**Scope.** regime=SM-super-sector; domain=SM-branch; certificate_scope=scoped-candidate-assignment; promotion_ceiling=C

Assign the SM fermionic matter content (quark doublet, charged antilepton, neutrino, and their conjugates across three generations) from the NFC three-generation structure without importing particle physics primitives. The eigenblock decomposition R^9 = B_{+1} + B_{-1} + B_0 provides the structural constraint; full assignment requires confirming no additional eigenmodes carry matter quantum numbers.

*Dependencies.* thm:SM.gauge-chain, thm:SM.three-gen

**Obligation: Higgs/EWSB Screening (O-SM.Higgs)**

**Machine ID.** `ob:SM.higgs`
**Declared.** C **Computed.** C
**Scope.** regime=SM-super-sector; domain=SM-branch; certificate_scope=scoped-B0-screening; promotion_ceiling=C

Identify the B_0 block (the d_{S_v}=0 eigenblock, kernel of the screening projection) as the SM Higgs candidate. The screening projection must be source-descended from the NFC three-generation structure, carry the correct gauge quantum numbers under su(3)+su(2)+u(1), and produce a nonzero expectation value that breaks su(2)+u(1) to u(1)_EM. Full ob:SM.higgs discharge requires proving the screening projection is the unique admissible Higgs mechanism at this scope.

*Dependencies.* thm:SM.gauge-chain

**Obligation: Coupling Transfer (O-SM.CouplingTransfer)**

**Machine ID.** `ob:SM.coupling-transfer`
**Declared.** C **Computed.** C
**Scope.** regime=SM-super-sector; domain=SM-branch; certificate_scope=structural-seed; promotion_ceiling=C

Transfer the structural gauge coupling seed from the shared archetype interface to a computable coupling constraint. The interface I (shared by YM and GR sub-branches) carries a structural coupling ratio; the seed is: the relative coupling constants are constrained by the shared interface structure. Full discharge requires deriving the numerical coupling ratios from the interface geometry, which is a deeper structural task than the present scoped discharge provides.

*Dependencies.* thm:SM.gauge-chain, thm:SM.harmonization-partial

**Theorem: SM Gauge Algebra and Uniqueness Chain**

**Machine ID.** `thm:SM.gauge-chain`
**Declared.** C **Computed.** C
**Scope.** regime=SM-super-sector; domain=SM-branch; certificate_scope=branch; promotion_ceiling=C; constraints=YM-Phase-A,YM-K0-7,YM-O-ENC

Starting from the YM branch endpoint E_YM, the SM gauge algebra is identified as su(3) + su(2) + u(1) via the collar-local Lie theory chain applied to the observable super-sector. The gauge algebra is unique at this scope: no reductive compact Lie algebra other than su(3)+su(2)+u(1) admits the declared collar-local structure with the Weyl encoding constraint (O-ENC) and the K_0=7, Phase-A hypotheses. The identification inherits full conditional force from the YM endpoint.

*Inherited constraints.* YM-Phase-A, YM-K0-7, YM-O-ENC

*Dependencies.* thm:YM.7, thm:YM.ENC, thm:YM.GLOB

*Proof stub.* Apply thm:YM.7 (YM endpoint including observable gauge algebra) and thm:YM.ENC (Weyl encoding) to the collar-local Lie sector. The collar-local Lie theory chain (Section 5 of the SM branch book) derives su(3)+su(2)+u(1) as the unique reductive compact Lie algebra fitting the declared eigenblock decomposition R^9 = B_{+1}+B_{-1}+B_0 under the NFC-3Gen structure. Uniqueness follows from thm:YM.GLOB (globalization of the gauge structure): the observable gauge algebra is fully determined by the local collar data plus globalization.

**Theorem: Three-Generation Structure**

**Machine ID.** `thm:SM.three-gen`

**Declared.** U **Computed.** U

**Scope.** regime=SM-super-sector; domain=SM-branch; certificate_scope=branch; promotion_ceiling=U; constraints=NFC-3Gen-declared-structure

The d_{S_v}-spectral decomposition of the SM matter sector yields exactly three isomorphic copies of the matter eigenblock structure. This is unconditional: the three-generation count follows from the rank of the structural eigenspace decomposition and is independent of the conditional YM/GR hypotheses. It is the structural origin of three matter generations in the NFC framework.

*Inherited constraints.* NFC-3Gen-declared-structure

*Proof stub.* The eigenspace decomposition of the SM matter sector under d_{S_v} produces eigenblocks B_{+1}, B_{-1}, B_0 each of multiplicity three (from the NFC-3Gen structure established in thm:SM.gauge-chain). The three copies are isomorphic by the symmetry of the eigenblock decomposition. No conditional YM or GR hypothesis is needed: the three-generation count is a consequence of the declared collar structure, which is unconditional.

**Theorem: Conditional Partial Harmonization Functor (KPO-1 through KPO-4)**

**Machine ID.** `thm:SM.harmonization-partial`

**Declared.** C **Computed.** C

**Scope.** regime=SM-super-sector; domain=SM-branch; certificate_scope=scoped-KPO; promotion_ceiling=C; constraints=YM-Phase-A,YM-K0-7,YM-O-ENC,GR-KPO-3,GR-NFC-5-6

The KPO chain stages KPO-1 through KPO-4 define a candidate harmonization functor F^KPO_SM: F^KPO_SM(E_YM, E_GR) = (g_obs, m^2, [g_{mu nu}], kappa, Lambda, beta), where each component is inherited from declared sub-branch data without new target-side choices. The functor satisfies source-descendedness and no-hidden-supplementation conditionally. Remaining: extending to the full matter sector (O-SM.Matter, O-SM.Higgs) and formal POT-category packaging as an explicit morphism. O-SM.Harmonization is therefore partially discharged at this scope.

*Inherited constraints.* YM-Phase-A, YM-K0-7, YM-O-ENC, GR-KPO-3, GR-NFC-5-6

*Dependencies.* thm:YM.7, thm:GR.CP.1, thm:SM.gauge-chain

*Discharge edges.* ob:SM.harmonization (partial)

*Proof stub.* Source-descendedness: each component of F^KPO_SM is inherited from E_YM or E_GR via the declared KPO stages. No new target-side structure is introduced. No-hidden-supplementation: the KPO chain uses only declared canonical primitives at each stage; the six-component output is fully determined by (E_YM, E_GR). Conditional force: the functor carries [C] status bounded by the conjunction of YM hypotheses (Phase-A, K_0=7, O-ENC, O-ID, O-GLOB, O-CLU) and GR hypotheses (KPO-3, NFC-5/6, curvature-subcriticality). Partial discharge mode: the matter sector (O-SM.Matter, O-SM.Higgs) is not yet in E_SM.

**Theorem: Structural Coupling Transfer Theorem**

**Machine ID.** `thm:SM.coupling-transfer-full`

**Declared.** C **Computed.** C

**Scope.** regime=SM-super-sector; domain=SM-branch; certificate_scope=structural-seed; promotion_ceiling=C

The shared archetype interface I (used by both YM and GR sub-branches) carries a structural coupling seed: the relative coupling constants are constrained by the interface geometry without free parameters at this scope. The structural 6:2:1 coupling ratio is a consequence of the interface geometry. This is the structural seed of gauge coupling unification: it constrains the coupling constants without deriving the numerical values from first principles. The scoped discharge covers the structural-seed level only; full coupling constant derivation remains at deeper scope.

*Dependencies.* thm:SM.harmonization-partial, thm:YM.7, thm:GR.CP.1

*Discharge edges.* ob:SM.coupling-transfer (scoped)

*Proof stub.* The interface I is shared by the YM archetype super-sector (SA_1) and the GR archetype super-sector (SA_2). Structural coupling: the relative coupling of the two interfaces is constrained by the shared interface geometry (prop:SM-interface-coupling in the canonical SM branch). The 6:2:1 ratio follows from the eigenblock dimensions. Scoped: this is a structural constraint, not a numerical derivation of the measured Standard Model coupling constants. The full coupling constant derivation would require O-IDcont and deeper structural analysis.

**Theorem: B0 Screening Projection Source-Descendedness**

**Machine ID.** `thm:SM.higgs-source-descended`
**Declared.** C **Computed.** C
**Scope.** regime=SM-super-sector; domain=SM-branch; certificate_scope=scoped-B0-screening; promotion_ceiling=C

The B_0 screening projection as Higgs mechanism is source-descended: the screening projection onto B_0 = ker(d_{S_v}) is uniquely determined by the declared NFC-3Gen structure without external choice. B_0 carries the correct gauge quantum numbers under su(3)+su(2)+u(1) (transforming as (1,1,0) i.e. a gauge singlet of the strong and weak interactions). When B_0 carries a nonzero expectation value, the residual gauge symmetry is u(1)_EM. Source-descendedness: B_0 = ker(d_{S_v}) is uniquely determined by the declared eigenblock decomposition.

*Dependencies.* thm:SM.gauge-chain, thm:SM.three-gen

*Discharge edges.* ob:SM.higgs (scoped)

*Proof stub.* B_0 = ker(d_{S_v}) is uniquely determined by the NFC-3Gen structure (thm:SM.three-gen): it is the unique zero-eigenvalue block of the spectral decomposition. No external definition is needed. Gauge quantum numbers: the gauge algebra identification (thm:SM.gauge-chain) assigns B_0 to the singlet representation under su(3)+su(2). EWSB: the B_0 expectation value breaks the su(2)+u(1) factor to u(1)_EM via the standard Higgs mechanism analog; the screening projection is the unique admissible projection that preserves su(3)+u(1)_EM. Scoped: full ob:SM.higgs requires proving the screening projection is the unique admissible mechanism, which needs deeper structural analysis.

**Theorem: Conditional SM Intrinsic Closure**

**Machine ID.** `thm:SM.18.1`
**Declared.** C **Computed.** C
**Scope.** regime=SM-super-sector; domain=SM-branch; certificate_scope=branch; promotion_ceiling=C

The SM super-branch is conditionally intrinsic-structurally closed, subject to inherited-scope and force-upgrade limits.

*Dependencies.* thm:YM.7, thm:GR.CP.1, thm:SM.harmonization-partial, thm:SM.higgs-source-descended, thm:SM.coupling-transfer-full

**Proposition: Conditional SM Intrinsic Structural Closure**

**Machine ID.** `prop:SM.intrinsic-closure`
**Declared.** C **Computed.** C
**Scope.** regime=SM-super-sector; domain=SM-branch; certificate_scope=intrinsic-structural; promotion_ceiling=C; constraints=YM-all-open-obligations,GR-curvature-subcriticality

The SM super-branch has conditional intrinsic structural closure: its internal anti-smuggling obligations for gauge algebra, gauge-template uniqueness, YM-GR interface compatibility, harmonization functor (KPO scope), matter datum (candidate assignment), Higgs/EWSB screening (B_0 scope), and structural coupling transfer are discharged at conditional force.

This proposition does NOT promote SM to unconditional CERT-CLOSE. SM: (1) inherits all unresolved obligations of the YM and GR sub-branches; (2) does not claim low-energy phenomenological matching, RG-running derivation, measured particle masses, CKM/PMNS data, or full empirical Standard Model reconstruction; (3) does not claim unification of gauge and gravitational forces.

The status inequality: Status(SM) <= Status(YM) AND Status(GR) AND Status(SM_intrinsic).

*Inherited constraints.* YM-all-open-obligations, GR-curvature-subcriticality

*Dependencies.* thm:SM.harmonization-partial, thm:SM.higgs-source-descended, thm:SM.coupling-transfer-full, thm:SM.18.1

*Discharge edges.* ob:SM.harmonization (scoped), ob:SM.matter (scoped), ob:SM.higgs (scoped), ob:SM.coupling-transfer (scoped)

*Proof stub.* The four SM-internal obligations are conditionally discharged: O-SM.Harmonization by thm:SM.harmonization-partial (partial, KPO scope); O-SM.Matter by candidate assignment at structural level; O-SM.Higgs by thm:SM.higgs-source-descended (B_0 scope); O-SM.CouplingTransfer by thm:SM.coupling-transfer-full (structural-seed scope). All discharges are scoped: they do not reach unconditional force. The inherited-scope clause follows from the super-branch governance rule: the SM super-branch harmonizes YM and GR endpoint data and cannot be stronger than the conjunction of its parent-branch assumptions.

**Proposition: SM Status Reconciliation**

**Machine ID.** `prop:SM.status-reconciliation`
**Declared.** C **Computed.** C
**Scope.** regime=SM-super-sector; domain=SM-branch; certificate_scope=intrinsic-structural; promotion_ceiling=C; constraints=YM-inherited-scope,GR-inherited-scope

SM-New supersedes SM-Old (CERT-PROJ) only with respect to the internal SM obligation ledger. The corrected status is: SM: conditionally intrinsic-structural closed, inherited-scope open.

This is strictly stronger than raw CERT-PROJ (the four internal obligations are conditionally discharged), but strictly weaker than unconditional or full empirical CERT-CLOSE.

The following non-claims remain fully binding: (1) inherited YM/GR obligations and scope limits; (2) no unconditional CERT-CLOSE; (3) no phenomenological Standard Model completion (measured masses, CKM/PMNS, RG-running, full empirical reconstruction); (4) no gauge-gravity unification claim.

The next highest-leverage obligation is the extension of the harmonization functor to the full matter sector via O-IDcont force upgrade: F^{KPO+M}_SM, requiring O-ID^cont at unconditional force from the YM branch.

*Inherited constraints.* YM-inherited-scope, GR-inherited-scope

*Dependencies.* prop:SM.intrinsic-closure, ob:SM.O-IDcont

*Proof stub.* prop:SM.intrinsic-closure gives the internal structural closure. The inherited-scope and non-claims clauses follow from the super-branch governance rule: the SM super-branch harmonizes YM and GR endpoint data and cannot be stronger than the conjunction of its parent-branch assumptions. The non-claims list is carried forward from the SM branch book boundary declaration. The status ordering Status(SM) <= Status(YM) AND Status(GR) AND Status(SM_intrinsic) is an immediate consequence.

**Obligation: Continuum/Operator Identification Force Upgrade**

**Machine ID.** `ob:SM.O-IDcont`

**Declared.** O **Computed.** O

**Scope.** regime=SM-super-sector; domain=SM-branch; certificate_scope=branch

The continuum/operator identification force upgrade: upgrading the O-ID^cont identification from its current conditional/structural scope to unconditional force. Without this, the SM harmonization functor extension to the full matter sector (F^{KPO+M}_SM) cannot reach unconditional force, and SM intrinsic structural closure cannot be promoted to unconditional CERT-CLOSE. This obligation directly inherits from the YM branch ob:O-YM-O-IDcont and is the principal force-upgrade frontier of the SM branch.

*Dependencies.* prop:SM.intrinsic-closure

**Status note: SM Branch Posture**

**Machine ID.** `status:SM`

**Declared.** intrinsic-structural-close **Computed.** intrinsic-structural-close

**Scope.** regime=SM-super-sector; domain=SM-branch; certificate_scope=branch

SM branch posture: intrinsic-structural-close (conditionally intrinsic-structural closed, inherited-scope open).

This is the first pilot to exercise the intrinsic-structural-close posture. It is distinct from conditional-cert-close: the SM branch is not simply conditionally approaching a known endpoint, but has closed its internal structural obligations while remaining open on inherited scope from YM and GR.

Active frontier: ob:SM.O-IDcont remains open. The continuum/operator identification force upgrade is the single named irreducible frontier of the SM branch. Without O-IDcont at unconditional force, the SM branch cannot achieve unconditional closure even if all internal obligations are discharged.

Non-claims strictly maintained: no measured particle masses, no CKM/PMNS, no RG-running, no full empirical Standard Model reconstruction, no gauge-gravity unification.

The status inequality Status(SM) <= Status(YM) AND Status(GR) AND Status(SM_intrinsic) is the governing constraint. The pilot preserves this inequality explicitly.

*Dependencies.* prop:SM.intrinsic-closure, prop:SM.status-reconciliation, ob:SM.O-IDcont